



# ORACLE

## Academy



# Java Foundations

1-1

## About This Course

**ORACLE**  
Academy



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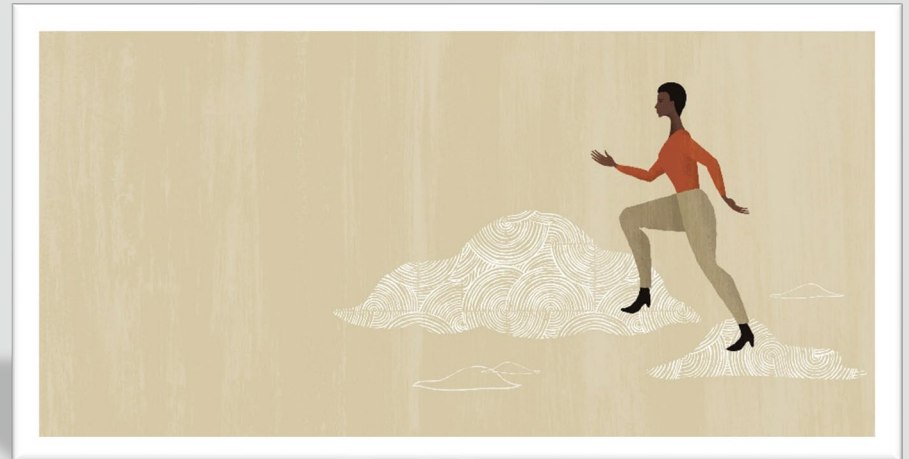
# Objectives

- This lesson covers the following objectives:
  - Identify course goals and objectives
  - Understand the course environment
  - Describe the course learning strategy



# Course Goals

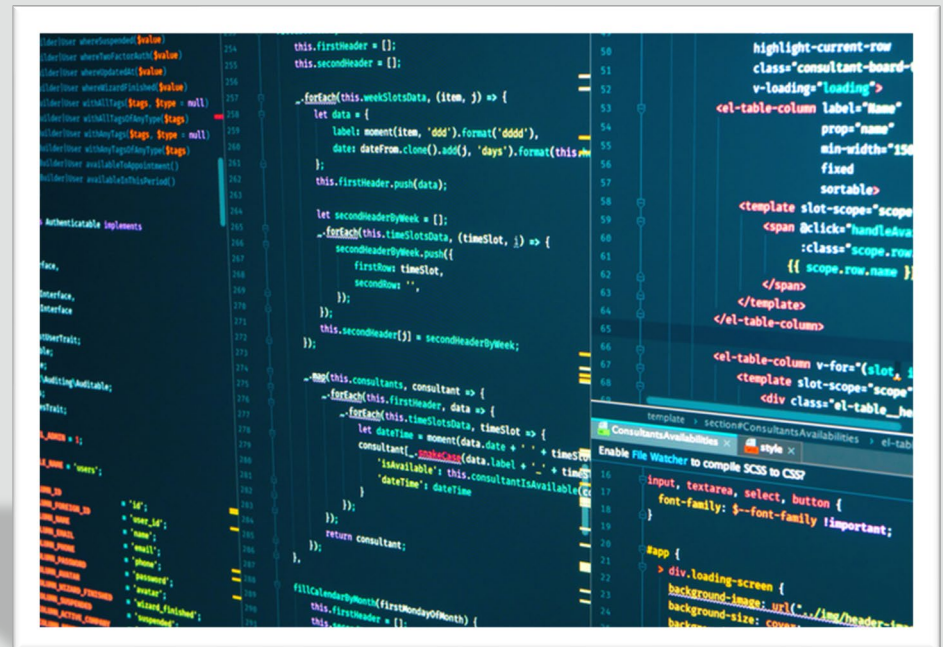
- Demonstrate knowledge of basic programming language concepts
- Demonstrate knowledge of the Java programming language
- Implement basic Java programming and object-oriented concepts





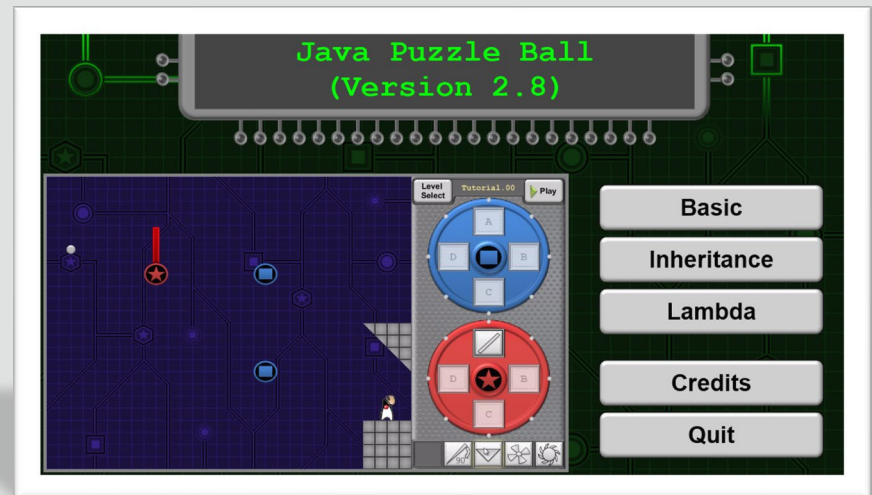


- A suitable Java Integrated Development Environment(IDE)
  - A tool used for writing code
  - Installation instructions are provided later



# Course Goals

- Java Puzzle Ball
  - A game used throughout the course
  - It's available through Oracle Academy Member Hub as soon as the relevant lessons occur
- Other materials are available from Oracle Academy Member Hub as the relevant lessons occur



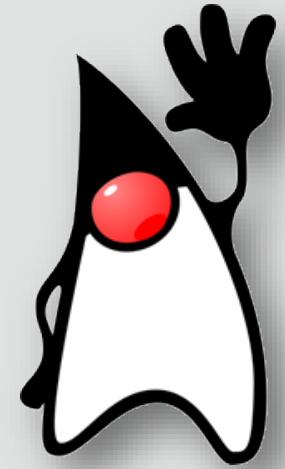
# Your Code

- You'll type your code in a Java IDE
- Using NetBeans, code looks like this:
  - The color of syntax highlighting will vary dependent on your chosen IDE

```
2 public class HelloWorld {  
3  
4     public static void main(String[] args) {  
5         System.out.println("Hello World!");  
6     }  
7 }
```

# Course Outline

- Section 1: What Is Java?
  - Lesson 1: About This Course
  - Lesson 2: Java: A Brief History
  - Lesson 3: Setting Up Java
- Section 2: Java Basics
  - Lesson 1: The Software Development Process
  - Lesson 2: What Is My Program Doing?
  - Lesson 3: Introduction to Object-Oriented Programming Concepts





# Course Outline

- Section 3: Data Types
  - Lesson 1: What Is a Variable?
  - Lesson 2: Numeric Data Types
  - Lesson 3: Textual Data Types
  - Lesson 4: Converting Between Data Types
  - Lesson 5: Keyboard Input



# Course Outline

- Section 4: Important Classes from the Java API
  - Lesson 1: What Is a Method?
  - Lesson 2: The `import` Declaration and Packages
  - Lesson 3: The `String` Class
  - Lesson 4: The `Random` Class
  - Lesson 5: The `Math` Class
- Section 5: Decision Statements
  - Lesson 1: Boolean Expressions and `if/else` Constructs
  - Lesson 2: Conditional Execution
  - Lesson 3: `switch` Statements

# Course Outline

- Section 6: Loop Statements
  - Lesson 1: `for` Loops
  - Lesson 2: `while` and `do/while` Loops
  - Lesson 3: `break` and `continue` Statements
- Section 7: Classes
  - Lesson 1: Creating a Class
  - Lesson 2: Instantiating Objects
  - Lesson 3: Overloading Methods
  - Lesson 4: Object Interaction and Encapsulation
  - Lesson 5: `static` Variables and Methods

# Course Outline

- Section 8: Arrays, ArrayLists, and Exceptions
  - Lesson 1: Arrays
  - Lesson 2: ArrayLists
  - Lesson 3: Exception Handling
  - Lesson 4: Debugging Concepts
- Section 9: GUI Applications with JavaFX
  - Lesson 1: Introduction to JavaFX
  - Lesson 2: Colors and Shapes
  - Lesson 3: Graphics, Audio, and Mouse Events
- Section 10: Final Project



# The Challenges of Designing This Course

- How do we design a Java course that will engage and educate a young audience?
  - Secondary schools
  - Two-year colleges
  - Four-year colleges
- How do we explain technical concepts to an audience that may have no computer science background?
  - This is an introductory programming course
  - It's designed to be taken with no prerequisite courses
  - But an understanding of basic algebra may be beneficial



# How We Won't Engage You

- Relying on slides that contain walls of text
- Bombarding you with information without giving you a chance to program
- Providing technobabble notes that don't explain or match the slides
- Forcing you to recall obscure technical facts



# How We'll Engage You

- In order for this course to connect with today's youth, its content must be more like you:
  - Hip
  - Trendy
  - Social
- We'll accomplish this with:
  - Backward hats
  - Sunglasses
  - Bling
  - Hashtags



# How We'll Really Engage You

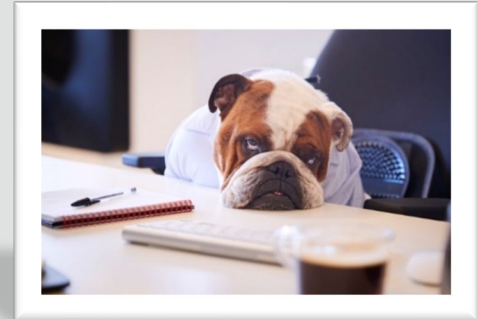
- The previous slide is satirical, but it also illustrates the conversational writing style of this course
- Lessons are written to prepare you to develop software, this is done by:
  - Illustrating core concepts, you'll need to create software
  - Providing code examples that you can reference and build from
  - Giving you a chance to program and discover solutions
  - Explaining why certain things are so
  - Maintaining continuity with scenarios throughout each section. Icons in the "Course Outline" slides hint at these scenarios
- We want you to feel good about your coding abilities

# Course Components

- Course components include:
  - Lessons
  - Small exercises
  - Quizzes
  - Practices
- All lesson materials should be available through Oracle Academy Member Hub
- This course also includes mid-term and final multiple-choice exams

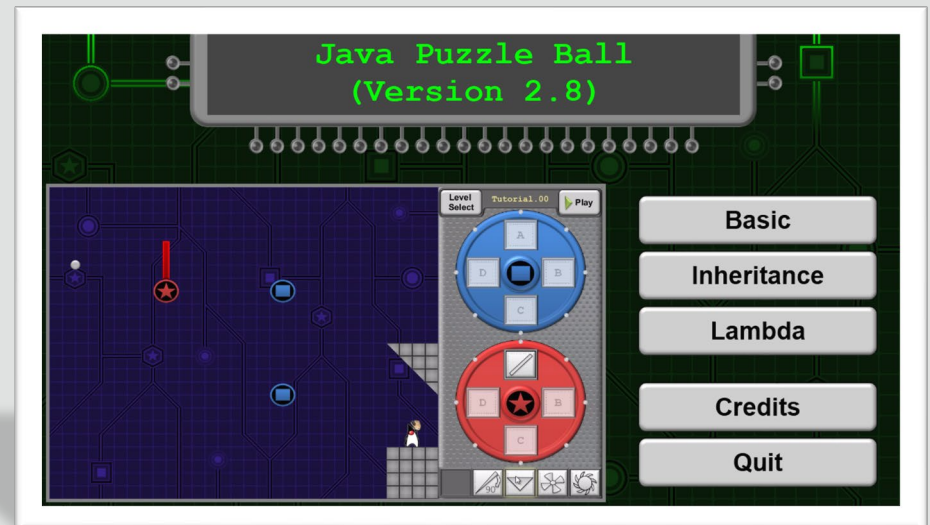
# The Problem with Long Lectures

- Students told us that long lectures made them feel exhausted, confused, and helpless when it was finally time to code
- Most learning occurred when students were allowed to code
  - The goal of this course is to be hands-on and project-based
  - That's why we include several smaller exercises throughout each lesson
  - We also found it counterproductive to bunch exercises at the end of each lesson



# Small Exercises

- Every lesson (starting with Lesson 1-3) includes small exercises that are designed to:
  - Establish familiarity with a concept
  - Build confidence as concepts are introduced
  - Build on existing knowledge
  - Allow you to experiment
- Exercises include:
  - Listing ideas
  - Playing games
  - Editing code



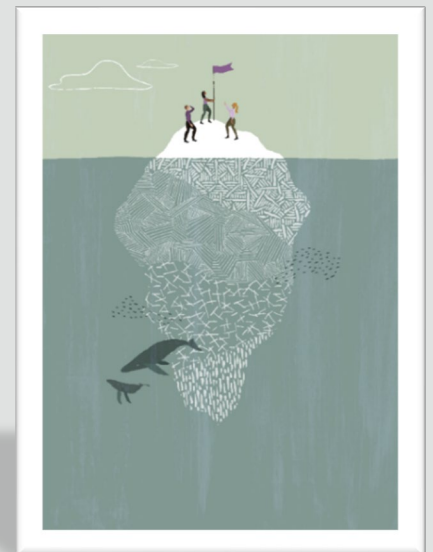


# Practices

- Every section includes a practice
- These are large exercises that should require several hours to complete
- They describe the features that a particular piece of software requires
- It's up to you to figure out how to implement these requirements
- Lessons are designed to explain everything you'll need to find a solution

# What You Could Do After Taking This Course

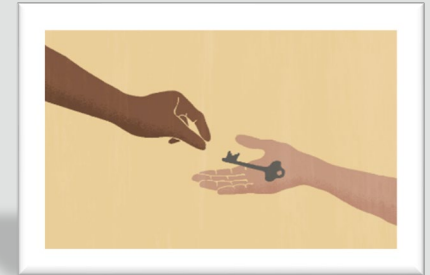
- Enjoy creating simple programs
- Appreciate an engineer's perspective
- Take more advanced courses:
  - Oracle Academy Java Fundamentals
  - Oracle Academy Java Programming
  - Oracle Certified Foundations Associate, Java
  - Advanced Placement Computer Science
- Explore the Oracle Academy Hub Career Center





# Certification

- Consider testing for an Oracle Certification
- Exam Number 1Z0-811
- Exam Name Java Foundations
  - Credential:
    - Java Certified Foundations Associate
  - Oracle Certification exams are multiple choice
  - Exams are managed by Oracle University, not Oracle Academy
  - Of all the certifications that Oracle offers, the Java Foundations exam most closely reflects the concepts you'll learn in this course, but the two don't align perfectly
  - You'll most likely need further studying to prepare for the certification exam



# Summary

- In this lesson, you should have learned how to:
  - Identify course goals and objectives
  - Understand the course environment
  - Describe the course learning strategy





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